

EXPERIMENT - 11

Title

Write an AngularJS Application to Access JSON File Data of an Employee from a Server using `$http` Service

Aim

To develop an AngularJS application that retrieves employee data from a JSON file hosted on a server using the `$http` service.

Prerequisites

- Knowledge of HTML, CSS, JavaScript
 - Basic understanding of AngularJS
 - Knowledge of JSON format
 - Node.js installed (to serve JSON file)
-

Software Requirements

- OS: Windows / Linux / macOS
- Browser: Google Chrome
- Code Editor: VS Code
- AngularJS CDN
- Node.js (for local server)

Theory

In **AngularJS**, the `$http` service is used to communicate with remote HTTP servers.

`$http` Service

- Sends `GET`, `POST`, `PUT`, `DELETE` requests
- Returns a promise
- Used to fetch JSON data

JSON (JavaScript Object Notation)

JSON is a lightweight data-interchange format used to store and transmit structured data.

Algorithm

1. Create a JSON file with employee data.
2. Create a Node.js server to serve JSON.
3. Create AngularJS application.
4. Use `$http.get()` method.
5. Display employee data in table format.
6. Run and verify output.

Project Structure

None

fsd_exp11/

|— server.js

|— employees.json

|— index.html

employees.json

JSON

```
[
  {
    "id": 1,
    "name": "Ravi Kumar",
    "department": "IT",
    "salary": 50000
  },
  {
    "id": 2,
    "name": "Anjali Sharma",
    "department": "HR",
    "salary": 45000
  },
  {
    "id": 3,
    "name": "Suresh Reddy",
    "department": "Finance",
    "salary": 60000
  }
]
```

server.js (Node.js Server)

JavaScript

```
const http = require('http');
const fs = require('fs');

const server = http.createServer((req, res) => {

  if (req.url === '/employees') {
    fs.readFile('employees.json', (err, data) => {
      if (err) {
        res.writeHead(500);
        res.end("Error loading JSON");
      } else {
        res.writeHead(200, { 'Content-Type':
'application/json' });
        res.end(data);
      }
    });
  }

  else {
    fs.readFile('index.html', (err, data) => {
      res.writeHead(200, { 'Content-Type':
'text/html' });
      res.end(data);
    });
  }

});

server.listen(3000, () => {
  console.log("Server running at http://localhost:3000");
});
```

index.html (AngularJS Application)

HTML

```
<!DOCTYPE html>
<html ng-app="empApp">
<head>
  <title>Employee Data</title>

  <script
src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.2/
angular.min.js"></script>

  <style>
    table {
      border-collapse: collapse;
      width: 60%;
      margin: auto;
    }

    th, td {
      border: 1px solid black;
      padding: 8px;
      text-align: center;
    }

    th {
      background-color: lightblue;
    }
  </style>
</head>
```

```

<body ng-controller="empController">

<h2 style="text-align:center;">Employee Details</h2>

<table>
  <tr>
    <th>ID</th>
    <th>Name</th>
    <th>Department</th>
    <th>Salary</th>
  </tr>

  <tr ng-repeat="emp in employees">
    <td>{{emp.id}}</td>
    <td>{{emp.name}}</td>
    <td>{{emp.department}}</td>
    <td>{{emp.salary}}</td>
  </tr>
</table>

<script>
  var app = angular.module("empApp", []);

  app.controller("empController", function($scope, $http)
  {

    $http.get("http://localhost:3000/employees")
      .then(function(response) {
        $scope.employees = response.data;
      }, function(error) {
        console.log("Error fetching data");
      });
  });

```

```
});  
</script>  
  
</body>  
</html>
```

Execution Steps

Step 1: Run Server

```
Shell  
node server.js
```

Step 2: Open Browser

```
None  
http://localhost:3000
```

Sample Output

ID	Name	Department	Salary
1	Ravi Kumar	IT	50000
2	Anjali Sharma	HR	45000
3	Suresh Reddy	Finance	60000

Expected Result

The AngularJS application successfully retrieves employee data from the server using `$http` service and displays it dynamically in a table.

Result

Thus, the AngularJS application to access JSON file data from a server using `$http` service was successfully implemented and executed.

Viva Voce Questions

1. What is `$http` service?
 2. What is JSON?
 3. What is `ng-repeat`?
 4. Why do we need a server to access JSON?
 5. What is promise in AngularJS?
 6. What happens if the server is not running?
-

Conclusion

This experiment demonstrates how AngularJS communicates with a backend server to fetch and display JSON data dynamically using the `$http` service.